



FAST NUCES

Department of Cyber Security

Computer Networks Project Report

Enterprise Network Design and Implementation Using

VLSM, OSPF, EIGRP, RIP, NAT and ACL

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1. Introduction

This report presents the design, configuration, and verification of a large-scale enterprise network implemented using Cisco Packet Tracer. The network was designed from scratch based on a given private IP address and a set of host requirements per subnet. The objective was to demonstrate a thorough understanding of Variable Length Subnet Masking (VLSM), multiple dynamic routing protocols, inter-protocol redistribution, Network Address Translation (NAT), Access Control Lists (ACL), and DHCP-based dynamic IP assignment.

The network consists of four major routing domains, namely OSPF Area 1, EIGRP Autonomous System 5, OSPF Area 2, and RIP Version 2, interconnected through boundary routers that perform mutual route redistribution. The design follows a hierarchical approach ensuring scalability, efficiency, and controlled access between different segments of the network.

2. Project Objectives

The main objectives of this project are as follows:

- Design a complete enterprise network topology using Cisco Packet Tracer Instructor Version.
- Apply Variable Length Subnet Masking (VLSM) to efficiently allocate IP addresses across all eleven networks (A through K).
- Configure OSPF Area 1, EIGRP AS 5, OSPF Area 2, and RIP v2 as the routing protocols for their respective domains.
- Implement route redistribution at the boundary routers connecting different routing domains.
- Configure a centralized DHCP server in Network K to provide dynamic IP addresses to all hosts across all routing domains.
- Implement Network Address Translation (NAT) on Router 20 for Network J using the provided public IP address.
- Configure Access Control Lists (ACL) on Router 16 to restrict specific hosts from accessing the Web Server in Network H.
- Set up a Mail Server in Network C and configure email communication for all hosts in OSPF Area 1.

3. Network Overview

The overall network is divided into four routing domains as described below. Each domain contains a set of LAN networks connected to routers, which are in turn interconnected through serial /30 WAN links.

3.1 Routing Domains

Routing Domain	Routers	Networks	Boundary Router
OSPF Area 1	R0, R3, R4, R5, R6	A, B, C	R5 (to EIGRP 5)
EIGRP AS 5	R5, R8, R9, R10, R11, R12, R13	D, E, F	R13 (to OSPF Area 2)
OSPF Area 2	R13, R14, R15, R16, R21	G, H, I	R21 (to RIP v2)
RIP Version 2	R17, R18, R19, R20, R21	J, K	R21 (from OSPF Area 2)

3.2 Network Topology Diagram

The following figure illustrates the complete network topology as implemented in Cisco Packet Tracer:

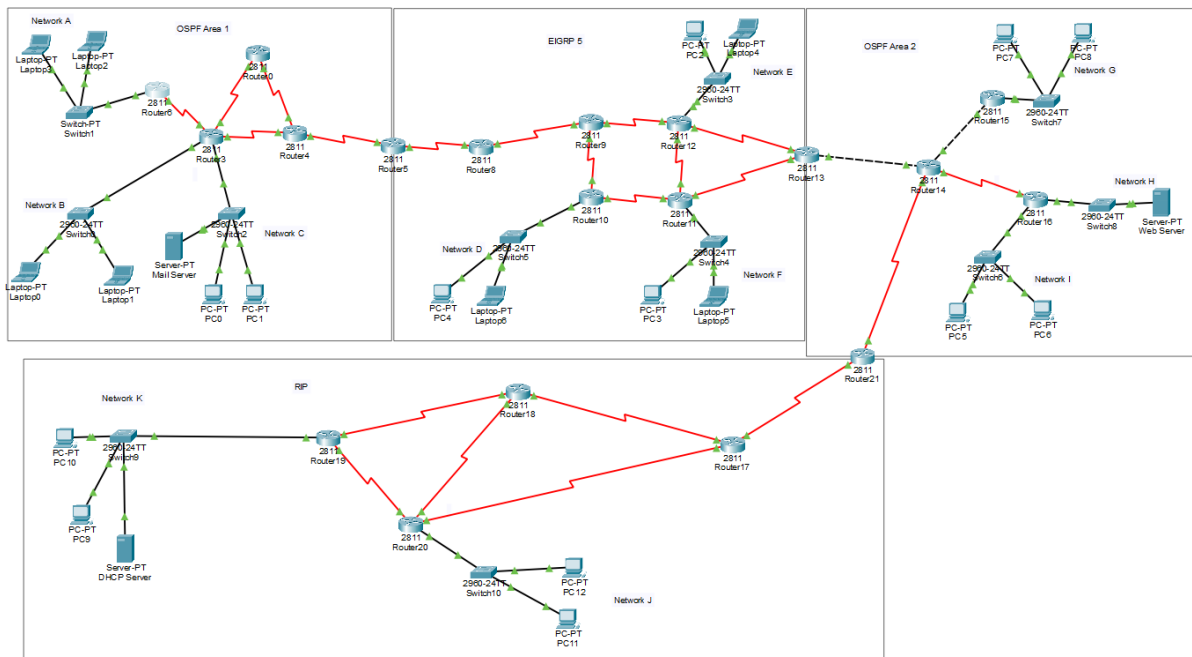


Figure 1: Complete Enterprise Network Topology

4. IP Addressing and VLSM Design

The private IP address assigned to this student is 27.128.22.25. Using the given host requirements for all eleven subnets (A through K), the total number of required hosts was calculated as follows:

4.1 CIDR Block Selection

Parameter	Value
Assigned Private IP	27.128.22.25
Total Required Hosts	612,229
$2^{19} - 2 = 524,286$	Insufficient ✗
$2^{20} - 2 = 1,048,574$	Sufficient ✓
Host Bits Required	20
Selected CIDR	/12 ($32 - 20 = 12$)
Base Network Block	27.128.0.0/12
Usable Range	27.128.0.0 – 27.143.255.255

4.2 Network Rank Order (Largest to Smallest)

According to VLSM methodology, networks are allocated from largest to smallest to avoid address wastage:

Rank	Network	Hosts Required	Host Bits (n)	CIDR	Block Size
1	G	90,123	17	/15	131,072
2	I	89,012	17	/15	131,072
3	B	78,901	17	/15	131,072
4	K	78,901	17	/15	131,072
5	D	67,890	17	/15 *	131,072
6	F	56,789	16	/16	65,536
7	H	45,678	16	/16	65,536
8	A	34,567	16	/16	65,536
9	J	34,567	16	/16	65,536
10	C	23,456	15	/17	32,768
11	E	12,345	14	/18	16,384

*** Note:** Network D requires 67,890 hosts. A /16 block only accommodates 65,534 usable hosts ($2^{16} - 2$), which is insufficient. Therefore, Network D is also allocated a /15 block, same as Networks G, I, B, and K.

4.3 LAN Subnet Allocation Table

Net	CIDR	Subnet Mask	Network Address	Gateway	Broadcast	Router
G	/15	255.254.0.0	27.128.0.0	27.128.0.1	27.129.255.255	R15
I	/15	255.254.0.0	27.130.0.0	27.130.0.1	27.131.255.255	R16
B	/15	255.254.0.0	27.132.0.0	27.132.0.1	27.133.255.255	R3
K	/15	255.254.0.0	27.134.0.0	27.134.0.1	27.135.255.255	R19
D	/15	255.254.0.0	27.136.0.0	27.136.0.1	27.137.255.255	R10
F	/16	255.255.0.0	27.138.0.0	27.138.0.1	27.138.255.255	R11
H	/16	255.255.0.0	27.139.0.0	27.139.0.1	27.139.255.255	R16
A	/16	255.255.0.0	27.140.0.0	27.140.0.1	27.140.255.255	R6
J	/16	255.255.0.0	27.141.0.0	27.141.0.1	27.141.255.255	R20
C	/17	255.255.128.0	27.142.0.0	27.142.0.1	27.142.127.255	R3
E	/18	255.255.192.0	27.142.128.0	27.142.128.1	27.142.191.255	R12

4.4 WAN Link (/30) Allocation

All point-to-point serial links between routers use /30 subnets (4 addresses: network, 2 hosts, broadcast). The WAN links are allocated sequentially starting from 27.142.192.0/30:

#	Link	Network /30	IP 1	Router	IP 2	Router
1	R6↔R3	27.142.192.0/30	27.142.192.1	R6	27.142.192.2	R3
2	R3↔R4	27.142.192.4/30	27.142.192.5	R3	27.142.192.6	R4
3	R3↔R0	27.142.192.8/30	27.142.192.9	R0	27.142.192.10	R3
4	R0↔R4	27.142.192.12/30	27.142.192.13	R0	27.142.192.14	R4
5	R4↔R5	27.142.192.16/30	27.142.192.17	R4	27.142.192.18	R5
6	R5↔R8	27.142.192.20/30	27.142.192.21	R5	27.142.192.22	R8
7	R8↔R9	27.142.192.24/30	27.142.192.25	R8	27.142.192.26	R9
8	R9↔R10	27.142.192.28/30	27.142.192.29	R9	27.142.192.30	R10
9	R9↔R12	27.142.192.32/30	27.142.192.33	R9	27.142.192.34	R12
10	R10↔R11	27.142.192.36/30	27.142.192.37	R10	27.142.192.38	R11
11	R11↔R12	27.142.192.40/30	27.142.192.41	R12	27.142.192.42	R11
12	R12↔R13	27.142.192.44/30	27.142.192.45	R12	27.142.192.46	R13
13	R11↔R13	27.142.192.48/30	27.142.192.49	R11	27.142.192.50	R13
14	R13↔R14	27.142.192.52/30	27.142.192.53	R13	27.142.192.54	R14
15	R14↔R15	27.142.192.56/30	27.142.192.57	R14	27.142.192.58	R15

#	Link	Network /30	IP 1	Router	IP 2	Router
16	R14↔R16	27.142.192.60/30	27.142.192.61	R14	27.142.192.62	R16
17	R14↔R21	27.142.192.64/30	27.142.192.65	R14	27.142.192.66	R21
18	R21↔R17	27.142.192.68/30	27.142.192.69	R17	27.142.192.70	R21
19	R17↔R18	27.142.192.72/30	27.142.192.73	R17	27.142.192.74	R18
20	R17↔R20	27.142.192.76/30	27.142.192.77	R17	27.142.192.78	R20
21	R20↔R19	27.142.192.80/30	27.142.192.81	R19	27.142.192.82	R20
22	R20↔R18	27.142.192.84/30	27.142.192.85	R18	27.142.192.86	R20
23	R18↔R19	27.142.192.88/30	27.142.192.89	R18	27.142.192.90	R19

5. Routing Protocol Configuration

Four different routing protocols are used in this network. Each routing domain is configured independently, and mutual redistribution is performed at the boundary routers to allow full network-wide reachability.

5.1 OSPF Area 1

Open Shortest Path First (OSPF) is a link-state routing protocol. OSPF Area 1 covers the first routing block consisting of Routers R0, R3, R4, R5, and R6. This area contains Networks A, B, and C.

Router	Role	Interfaces in OSPF Area 1	Connected LAN
R0	Core	S1/0 (to R3), S1/1 (to R4)	None
R3	LAN Gateway	Fa0/0 (Net B), Fa0/1 (Net C), S1/0–3	Network B, C
R4	Core	S1/0 (to R3), S1/1 (to R0), S1/2 (to R5)	None
R5	Boundary	S1/0 (to R4) — OSPF side	None (redistribution)
R6	LAN Gateway	Fa0/0 (Net A), S1/0 (to R3)	Network A

The OSPF process uses Process ID 1 for all routers in this domain. All interfaces are configured under Area 1. Router R5 also runs EIGRP 5 and performs redistribution between OSPF Area 1 and EIGRP 5.

```

R3>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    27.0.0.0/8 is variably subnetted, 39 subnets, 6 masks
O E2   27.128.0.0/15 [110/20] via 27.142.192.6, 01:00:27, Serial1/2
O E2   27.130.0.0/15 [110/20] via 27.142.192.6, 01:00:27, Serial1/2
C       27.132.0.0/15 is directly connected, FastEthernet0/0
L       27.132.0.1/32 is directly connected, FastEthernet0/0
O E2   27.134.0.0/15 [110/20] via 27.142.192.6, 01:00:27, Serial1/2
O E2   27.136.0.0/15 [110/20] via 27.142.192.6, 01:00:52, Serial1/2
O E2   27.138.0.0/16 [110/20] via 27.142.192.6, 01:00:52, Serial1/2
O E2   27.139.0.0/16 [110/20] via 27.142.192.6, 01:00:27, Serial1/2
O       27.140.0.0/16 [110/65] via 27.142.192.1, 01:01:02, Serial1/0
O E2   27.141.0.0/16 [110/20] via 27.142.192.6, 01:00:27, Serial1/2
C       27.142.0.0/17 is directly connected, FastEthernet0/1

R3>

```

Figure 2: OSPF Routing Table on R3

5.2 EIGRP Autonomous System 5

Enhanced Interior Gateway Routing Protocol (EIGRP) is a Cisco proprietary advanced distance-vector routing protocol. EIGRP AS 5 covers the middle routing block consisting of Routers R5, R8, R9, R10, R11, R12, and R13. This block contains Networks D, E, and F.

Router	Role	Connected LAN
R5	Boundary (OSPF1↔EIGRP)	None
R8	Core	None
R9	Core	None
R10	LAN Gateway	Network D
R11	LAN Gateway	Network F
R12	LAN Gateway	Network E
R13	Boundary (EIGRP↔OSPF2)	None

```

R9>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    27.0.0.0/8 is variably subnetted, 37 subnets, 6 masks
D EX   27.128.0.0/15 [170/3193856] via 27.142.192.34, 01:01:49, Serial1/1
D EX   27.130.0.0/15 [170/3193856] via 27.142.192.34, 01:01:49, Serial1/1
D EX   27.132.0.0/15 [170/3193856] via 27.142.192.25, 01:02:14, Serial1/0
D EX   27.134.0.0/15 [170/3193856] via 27.142.192.34, 01:01:49, Serial1/1
D      27.136.0.0/15 [90/2172416] via 27.142.192.30, 01:02:34, Serial1/2
D      27.138.0.0/16 [90/2684416] via 27.142.192.34, 01:02:30, Serial1/1
           [90/2684416] via 27.142.192.30, 01:02:30, Serial1/2
D EX   27.139.0.0/16 [170/3193856] via 27.142.192.34, 01:01:49, Serial1/1
D EX   27.140.0.0/16 [170/3193856] via 27.142.192.25, 01:02:14, Serial1/0
D EX   27.141.0.0/16 [170/3193856] via 27.142.192.34, 01:01:49, Serial1/1
D EX   27.142.0.0/17 [170/3193856] via 27.142.192.25, 01:02:14, Serial1/0
--More-- |

```

Figure 3: EIGRP Routing Table on R9

5.3 OSPF Area 2

A second OSPF process (Process ID 2) is used for the third routing block. This area consists of Routers R13, R14, R15, R16, and R21, and contains Networks G, H, and I.

Router	Role	Connected LAN
R13	Boundary (EIGRP↔OSPF2)	None
R14	Core	None
R15	LAN Gateway	Network G
R16	LAN Gateway	Network H (Web Server) + Network I
R21	Boundary (OSPF2↔RIP)	None


```

R14>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    27.0.0.0/8 is variably subnetted, 38 subnets, 6 masks
O       27.128.0.0/15 [110/2] via 27.142.192.58, 01:02:38, FastEthernet0/1
O       27.130.0.0/15 [110/65] via 27.142.192.62, 01:03:13, Serial1/0
O E2    27.132.0.0/15 [110/20] via 27.142.192.53, 01:02:38, FastEthernet0/0
O E2    27.134.0.0/15 [110/20] via 27.142.192.66, 01:03:13, Serial1/1
O E2    27.136.0.0/15 [110/20] via 27.142.192.53, 01:02:38, FastEthernet0/0
O E2    27.138.0.0/16 [110/20] via 27.142.192.53, 01:02:38, FastEthernet0/0
O       27.139.0.0/16 [110/65] via 27.142.192.62, 01:03:13, Serial1/0
O E2    27.140.0.0/16 [110/20] via 27.142.192.53, 01:02:38, FastEthernet0/0
O E2    27.141.0.0/16 [110/20] via 27.142.192.66, 01:03:13, Serial1/1
O E2    27.142.0.0/17 [110/20] via 27.142.192.53, 01:02:38, FastEthernet0/0
O E2    27.142.128.0/18 [110/20] via 27.142.192.53, 01:02:38, FastEthernet0/0
--More--

```

Figure 4: OSPF Area 2 Routing Table on R14

5.4 RIP Version 2

Routing Information Protocol Version 2 (RIP v2) is used in the fourth and final routing block. It covers Routers R17, R18, R19, R20, and R21, containing Networks J and K. RIP v2 supports classless routing with the no auto-summary command enabled.

Router	Role	Connected LAN
R17	Core	None
R18	Core	None
R19	LAN Gateway	Network K (DHCP Server)
R20	LAN Gateway + NAT	Network J
R21	Boundary (OSPF2↔RIP)	None

```
R20>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    27.0.0.0/8 is variably subnetted, 38 subnets, 6 masks
R       27.128.0.0/15 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
R       27.130.0.0/15 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
R       27.132.0.0/15 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
R       27.134.0.0/15 [120/1] via 27.142.192.81, 00:00:20, Serial1/1
R       27.136.0.0/15 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
R       27.138.0.0/16 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
R       27.139.0.0/16 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
R       27.140.0.0/16 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
C       27.141.0.0/16 is directly connected, FastEthernet0/0
L       27.141.0.1/32 is directly connected, FastEthernet0/0
R       27.142.0.0/17 [120/6] via 27.142.192.77, 00:00:18, Serial1/0
--More--
```

Figure 5: RIP Routing Table on R20

6. Route Redistribution

Route redistribution is configured at the three boundary routers that connect different routing domains. Without redistribution, routes learned by one protocol would not be visible to routers running a different protocol, resulting in network isolation.

6.1 Boundary Routers

Router	Connects	Redistribution Direction
R5	OSPF Area 1 ↔ EIGRP AS 5	Redistributes OSPF into EIGRP and EIGRP into OSPF
R13	EIGRP AS 5 ↔ OSPF Area 2	Redistributes EIGRP into OSPF and OSPF into EIGRP
R21	OSPF Area 2 ↔ RIP v2	Redistributes OSPF into RIP and RIP into OSPF

On R5, the EIGRP redistribution into OSPF uses the subnets keyword to include subnetted routes. For EIGRP, a seed metric of 1544 2000 255 1 1500 is specified when redistributing OSPF routes, representing bandwidth, delay, reliability, load, and MTU respectively. The same metric parameters are applied on R13.

```
R15>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    27.0.0.0/8 is variably subnetted, 36 subnets, 6 masks
C       27.128.0.0/15 is directly connected, FastEthernet0/1
L       27.128.0.1/32 is directly connected, FastEthernet0/1
O       27.130.0.0/15 [110/66] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.132.0.0/15 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.134.0.0/15 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.136.0.0/15 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.138.0.0/16 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
O       27.139.0.0/16 [110/66] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.140.0.0/16 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.141.0.0/16 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
O E2    27.142.0.0/17 [110/20] via 27.142.192.57, 01:04:09, FastEthernet0/0
--More--
```

Figure 6: Route Redistribution Verification on R5

7. DHCP Server Configuration

A centralized DHCP server is located at IP address 27.134.0.2 within Network K (connected to Router R19). This server provides dynamic IP address assignment to hosts across Networks A, B, C, D, E, F, J, and K. Networks G and I use static IP addressing and therefore do not have DHCP pools configured.

7.1 DHCP Server Static IP Settings

Parameter	Value
Server IP Address	27.134.0.2
Subnet Mask	255.254.0.0
Default Gateway	27.134.0.1 (R19 Fa0/0)
DNS Server	27.134.0.2 (self)

7.2 ip helper-address

Since DHCP uses broadcast and broadcasts do not cross router boundaries, the ip helper-address command is configured on each router interface facing a LAN. This causes the router to forward DHCP broadcast requests as unicast packets to the DHCP server at 27.134.0.2.

7.3 DHCP Pools Configured on the Server

A total of 8 DHCP pools are configured. Each pool is limited to a maximum of 512 users as set in Packet Tracer. Networks G and I do not have pools as they use static addressing.

Pool Name	Default Gateway	DNS Server	Start IP	Subnet Mask	Max Users	Network
POOL_A	27.140.0.1	27.134.0.2	27.140.0.10	255.255.0.0	512	A /16
POOL_B	27.132.0.1	27.134.0.2	27.132.0.10	255.254.0.0	512	B /15
POOL_C	27.142.0.1	27.134.0.2	27.142.0.10	255.255.128.0	512	C /17
POOL_D	27.136.0.1	27.134.0.2	27.136.0.10	255.254.0.0	512	D /15
POOL_E	27.142.128.1	27.134.0.2	27.142.128.10	255.255.192.0	512	E /18
POOL_F	27.138.0.1	27.134.0.2	27.138.0.10	255.255.0.0	512	F /16
POOL_J	27.141.0.1	27.134.0.2	27.141.0.10	255.255.0.0	512	J /16
POOL_K	27.134.0.1	27.134.0.2	27.134.0.10	255.254.0.0	512	K /15

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	512	0.0.0.0	0.0.0.0
POOL_K	27.134.0.1	27.134.0.2	27.134.0.10	255.254.0.0	512	0.0.0.0	0.0.0.0
POOL_J	27.141.0.1	27.134.0.2	27.141.0.10	255.255.0.0	512	0.0.0.0	0.0.0.0
POOL_F	27.138.0.1	27.134.0.2	27.138.0.10	255.255.0.0	512	0.0.0.0	0.0.0.0
POOL_E	27.142.128.1	27.134.0.2	27.142.128.10	255.255.192.0	512	0.0.0.0	0.0.0.0
POOL_D	27.136.0.1	27.134.0.2	27.136.0.10	255.254.0.0	512	0.0.0.0	0.0.0.0
POOL_C	27.142.0.1	27.134.0.2	27.142.0.10	255.255.128.0	512	0.0.0.0	0.0.0.0
POOL_B	27.132.0.1	27.134.0.2	27.132.0.10	255.254.0.0	512	0.0.0.0	0.0.0.0
POOL_A	27.140.0.1	27.134.0.2	27.140.0.10	255.255.0.0	512	0.0.0.0	0.0.0.0

Figure 7: DHCP Server Pool Configuration

IP Configuration

☒ DHCP
 ☐ Static

IPv4 Address: 27.134.0.10
 Subnet Mask: 255.254.0.0
 Default Gateway: 27.134.0.1
 DNS Server: 27.134.0.2

Figure 8: Host Receiving Dynamic IP Address from DHCP Server

8. Network Address Translation (NAT)

Network Address Translation (NAT) is implemented on Router R20 for Network J. NAT allows the private IP addresses of hosts in Network J to be translated to a single public IP address when communicating with external networks. This is a common real-world technique to conserve public IP address space and provide a security boundary.

8.1 NAT Configuration Details

Parameter	Value
NAT Router	R20
Inside Network	Network J — 27.141.0.0/16
Public IP (Pool)	198.244.120.212
NAT Type	PAT (Port Address Translation / Overload)
Inside Interface	FastEthernet0/0 (ip nat inside)
Outside Interfaces	Loopback0, Serial1/0, Serial1/1, Serial1/2 (ip nat outside)
ACL Permitting NAT	access-list 1 permit 27.141.0.0 0.0.255.255

The public IP address 198.244.120.212 is configured on a Loopback interface on R20 and redistributed into RIP using the redistribute connected metric 1 command. This ensures that all routers across all domains have a return route to 198.244.120.212 so that replies to NATted packets can reach R20 and be de-translated.

```
R20>show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 198.244.120.212:7 27.141.0.11:7     27.140.0.3:7      27.140.0.3:7
icmp 198.244.120.212:8 27.141.0.11:8     27.140.0.3:8      27.140.0.3:8

R20>
```

Figure 9: Active NAT Translations on R20

```
C:\>ping 27.140.0.3

Pinging 27.140.0.3 with 32 bytes of data:

Reply from 27.140.0.3: bytes=32 time=11ms TTL=116
Reply from 27.140.0.3: bytes=32 time=10ms TTL=116
Reply from 27.140.0.3: bytes=32 time=53ms TTL=116
Reply from 27.140.0.3: bytes=32 time=76ms TTL=116

Ping statistics for 27.140.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 76ms, Average = 37ms

C:\>
```

Figure 10: End-to-End Ping Verification Across NAT Boundary

9. Access Control List (ACL)

An extended Access Control List (ACL) is configured on Router R16 to restrict access to the Web Server located in Network H. The ACL is applied outbound on the FastEthernet0/0 interface facing Network H, ensuring that matching traffic is dropped before it reaches the web server.

9.1 ACL Rules

Rule	Action	Source	Destination	Reason
1	DENY	Host 27.140.0.3 (Net A)	Web Server 27.139.0.2	Specific host in Network A blocked
2	DENY	27.136.0.0/15 (All Net D)	Web Server 27.139.0.2	All hosts in Network D blocked

Rule	Action	Source	Destination	Reason
3	PERMIT	Any	Any	All other traffic allowed

The extended ACL number 100 is applied to the outbound direction on R16 Fa0/0 using the command `ip access-group 100 out`. Using an outbound extended ACL is efficient because R16 can drop the packet as close to the destination as possible.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 27.139.0.2

Pinging 27.139.0.2 with 32 bytes of data:

Reply from 27.142.192.62: Destination host unreachable.
Reply from 27.142.192.62: Destination host unreachable.
Reply from 27.142.192.62: Destination host unreachable.
Reply from 27.142.192.62: Destination host unreachable.

Ping statistics for 27.139.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 11: ACL Blocking Specific Host from Web Server

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 27.139.0.2

Pinging 27.139.0.2 with 32 bytes of data:

Request timed out.
Reply from 27.139.0.2: bytes=32 time=7ms TTL=119
Reply from 27.139.0.2: bytes=32 time=57ms TTL=119
Reply from 27.139.0.2: bytes=32 time=61ms TTL=119

Ping statistics for 27.139.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 61ms, Average = 41ms

C:\>
```

Figure 12: ACL Permitting Normal Hosts to Access Web Server

10. Mail Server Configuration

A Mail Server is deployed in Network C with a static IP address of 27.142.0.2. The mail server uses SMTP (Simple Mail Transfer Protocol) for sending and POP3 for receiving email. All hosts within OSPF Area 1, which includes Networks A, B, and C, are configured to send and receive email through this mail server.

10.1 Mail Server Settings

Parameter	Value
Server IP	27.142.0.2
Subnet Mask	255.255.128.0
Default Gateway	27.142.0.1 (R3 Fa0/1)
Domain Name	mail.net
SMTP Service	Enabled
POP3 Service	Enabled
Incoming Mail Server	27.142.0.2
Outgoing Mail Server	27.142.0.2

10.2 User Accounts Created on Mail Server

Username	Email Address	Network
laptop0	laptop0@mail.net	Network A
laptop1	laptop1@mail.net	Network A
laptop2	laptop2@mail.net	Network A
laptop3	laptop3@mail.net	Network A
pc0	pc0@mail.net	Network C
pc1	pc1@mail.net	Network C

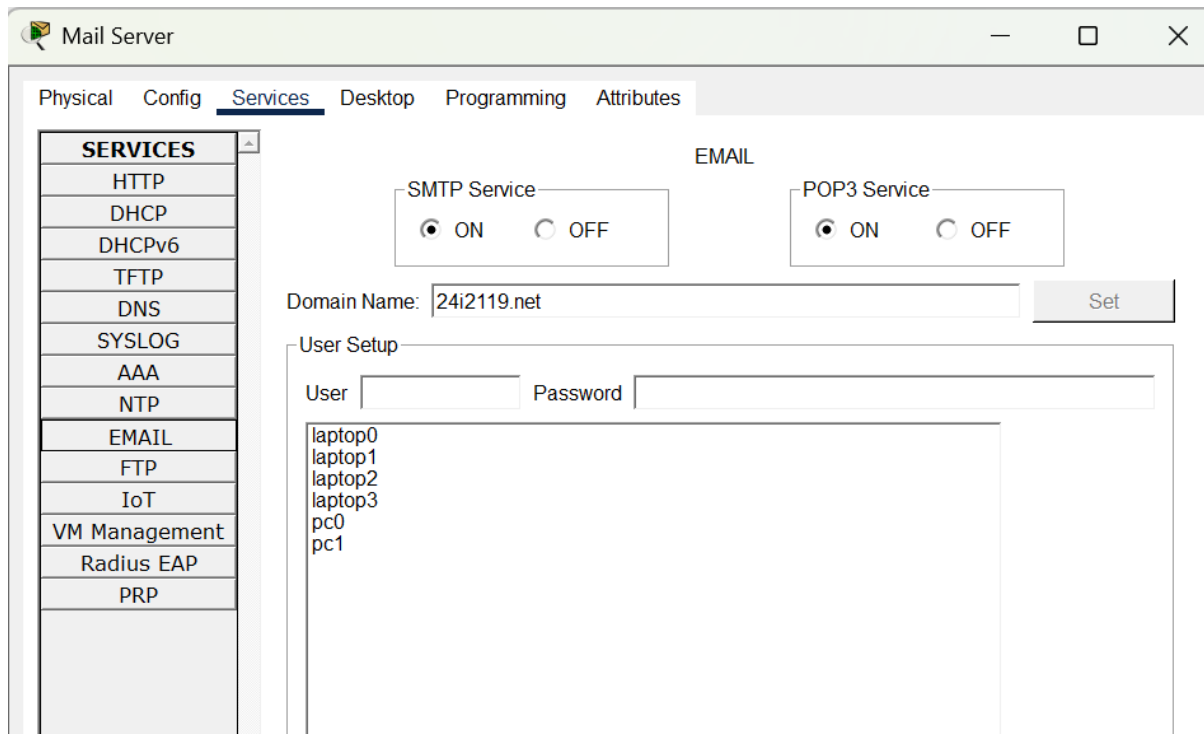


Figure 13: Mail Server Configuration in Packet Tracer

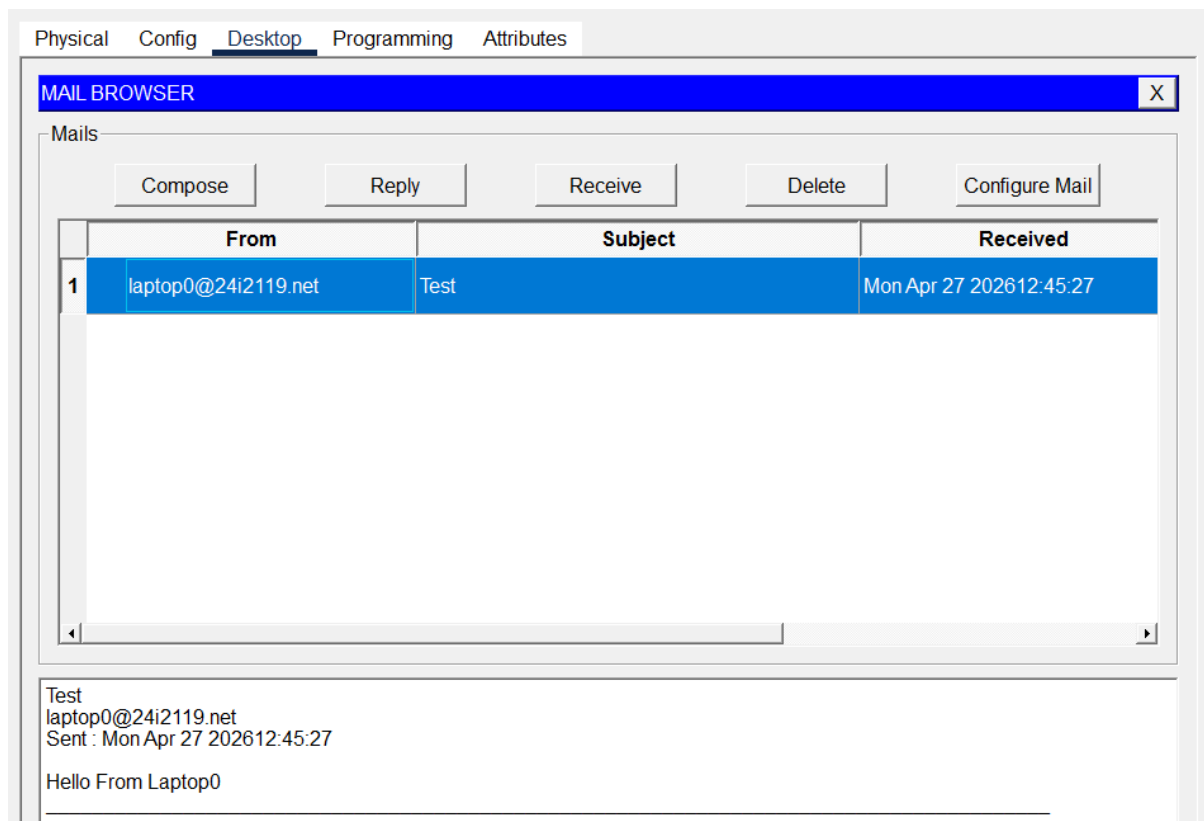


Figure 14: Email Communication Between Hosts in OSPF Area 1

11. Testing and Verification

After completing all configurations, extensive testing was performed to verify the correct operation of all network components. The following subsections summarize the test results.

11.1 Connectivity Test Summary

Test	Source	Destination	Protocol	Expected	Result
1	PC12 (Net J)	Laptop2 (Net A)	ICMP	Success (via NAT)	PASS
2	Laptop2 (Net A)	PC12 (Net J)	ICMP	Success	PASS
3	Host (Net D)	Web Server (H)	ICMP	BLOCKED by ACL	PASS
4	27.140.0.3 (A)	Web Server (H)	ICMP	BLOCKED by ACL	PASS
5	Host (Net B)	Web Server (H)	ICMP	Success	PASS
6	Host (Net G)	Host (Net K)	ICMP	Success	PASS
7	DHCP Client	DHCP Server	DHCP	IP Assigned	PASS
8	Laptop0	Laptop2	Email	Received	PASS

11.2 NAT Verification

The command `show ip nat translations` was executed on R20 to verify that PAT translations were being created correctly for outbound traffic from Network J. The translation table confirmed that private IP addresses from the 27.141.0.0/16 range were being mapped to the public IP 198.244.120.212 with unique port numbers.

```
R20>show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 198.244.120.212:7  27.141.0.11:7    27.140.0.3:7      198.244.120.212:7
icmp 198.244.120.212:8 27.141.0.11:8    27.140.0.3:8      198.244.120.212:8
R20>
```

Figure 15: Verified NAT Translation Table

11.3 Route Redistribution Verification

The `show ip route` command was executed on representative routers across all four routing domains. In each case, routes from all other domains were visible in the routing table, confirming that redistribution was functioning correctly at all three boundary routers.

```
R6>enable
R6#show ip route 198.244.120.212
Routing entry for 198.244.120.212/32
Known via "ospf 1", distance 110, metric 20, type intra area
  Last update from 27.142.192.2 on Serial1/0, 01:41:07 ago
  Routing Descriptor Blocks:
    * 27.142.192.2, from 27.142.192.21, 01:41:07 ago, via Serial1/0
      Route metric is 20, traffic share count is 1
R6#
```

Figure 16: R6 Routing Table Showing Redistributed NAT Public IP Route

12. Server IP Address Summary

Device	IP Address	Network	Subnet Mask	Gateway	Purpose
DHCP/DNS Server	27.134.0.2	K	255.254.0.0	27.134.0.1	Dynamic IP assignment + DNS
Mail Server	27.142.0.2	C	255.255.128.0	27.142.0.1	SMTP/POP3 email for OSPF Area 1
Web Server	27.139.0.2	H	255.255.0.0	27.139.0.1	HTTP service (ACL protected)
Blocked Host (Net A)	27.140.0.3	A	255.255.0.0	27.140.0.1	Denied access to Web Server
Blocked Range (Net D)	27.136.0.0 – 27.137.255.255	D	255.254.0.0	—	Entire Net D denied to Web Server

13. Conclusion

This project successfully demonstrates the design and implementation of a fully functional enterprise-grade network using Cisco Packet Tracer. The network spans four distinct routing domains — OSPF Area 1, EIGRP AS 5, OSPF Area 2, and RIP Version 2 — and employs route redistribution at three boundary routers to achieve complete network-wide reachability.

Variable Length Subnet Masking was applied rigorously to efficiently utilize the allocated /12 address block (27.128.0.0/12), ensuring that each network segment received exactly the right block size to accommodate its host requirements without significant waste. The /30 subnets were used for all point-to-point WAN links, following best practice for serial link addressing.

The centralized DHCP server in Network K, combined with the ip helper-address configuration on all LAN-facing router interfaces, ensures seamless dynamic IP address assignment across all routing domains. NAT with PAT on R20 allows the private hosts of Network J to communicate with the outside world using a single registered public IP address. The extended

ACL on R16 enforces the required security policy by blocking specific hosts from accessing the web server.

All configurations were tested and verified through ping tests, show ip route, show ip nat translations, and email communication tests, all of which produced the expected results, confirming the correctness and completeness of the implementation.

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